# Software Requirements Specification (SRS)

## Trade Track (Offline Trading Journal & Analytics Platform)

**Document Version:** 1.0.0 **Status:** Verified & Released **Target Application:** Trade Track **Platform:** Flutter (Android, iOS, Desktop)

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## 1. Introduction

### 1.1 Purpose

This Software Requirements Specification (SRS) document provides a complete description of the system architecture, functional requirements, data models, non-functional requirements, and operating constraints for **Trade Track**. It serves as the formal technical baseline for developers, product managers, and software testers.

### 1.2 Scope

**Trade Track** is an offline-first, private financial journal engineered for equity, commodity (MCX/NSE/BSE), and forex traders. The application records trading executions, computes mathematical and psychological metrics in real time (Gross/Net P&L, Risk-to-Reward ratio, R-Multiple, Expectancy, Maximum Drawdown), presents dynamic interactive visualizations, and enables daily performance retrospectives with calendar heatmaps.

### 1.3 Key Definitions & Acronyms

| Term | Definition |
|---|---|
| **Gross P&L** | Profit or Loss calculated as `(Exit Price - Entry Price) × Quantity × Multiplier` before fees. |
| **Net P&L** | `Gross P&L - Total Costs` (Brokerage + Exchange Fees + Spread + Other Fees). |
| **R:R** | Planned Risk-to-Reward Ratio: `Planned Reward / Planned Risk`. |
| **R-Multiple** | Realized reward multiple relative to initial dollar risk: `Net P&L / Initial Planned Risk`. |
| **Expectancy** | `(Win Rate × Avg Win) - (Loss Rate × Avg Loss)`. Expected monetary gain per trade. |
| **Max Drawdown** | The maximum peak-to-trough decline in account balance during a specific evaluation period. |
| **Drift** | A type-safe reactive persistence library for SQLite in Flutter. |
| **Riverpod** | Compile-safe, testable state management framework for Flutter. |

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# 2. Overall Description

## 2.1 Product Perspective

Trade Track is a standalone, client-side application. Unlike cloud-dependent trading platforms, all database records, chart assets, and analytical computations execute entirely on the local device. There is zero external server communication, eliminating latency and guaranteeing complete financial privacy.

```
graph TD
    UI[Presentation Layer - Flutter Widgets & GoRouter]
    Providers[State Layer - Riverpod Providers & Notifiers]
    Engine[Domain Layer - TradingCalculationEngine]
    Repos[Data Layer - Repository Pattern]
    DB[(Storage - Drift SQLite Database)]
    Disk[File Storage - Screenshots & CSV/JSON Backups]

    UI <--> Providers
    Providers --> Engine
    Providers <--> Repos
    Repos <--> DB
    Repos <--> Disk
```

## 2.2 System Architecture & Technology Stack

- **Framework**: Flutter 3.47+ (Dart 3.x)
- **Design System**: Luxury Onyx & Gold dark theme (`AppColors`, Material 3 design tokens)
- **State Management**: `flutter_riverpod` (v2.6.1)
- **Local Persistence**: `drift` (v2.35.2) with `sqlite3_flutter_libs`
- **Routing**: `go_router` (v18.1.0) with `StatefulShellRoute.indexedStack`
- **Data Visualizations**: `fl_chart` (v1.2.0)
- **Shared Preferences**: `shared_preferences` (v2.5.6)
- **System Utilities**: `share_plus`, `file_picker`, `image_picker`, `intl`, `uuid`, `csv`

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# 3. Current Implementation & Status

| Phase | Milestone | Current Status | Notes |
|---|---|---|---|
| **Core Architecture** | Repository, Engine, Database | □ Complete | In-memory & SQLite tests passing (100%) |
| **Static Analysis** | Linting & Linter Rules | □ 0 Issues | `flutter analyze` clean with 0 warnings |
| **Test Suite** | Unit, Widget & Integration | □ 24/24 Passed | 100% test pass rate across engine & db |
| **Android Packaging** | Release APK Assembly | □ Verified | Release APK generated (`66.4 MB`) |
| **Hardware Deployment** | Physical Device Installation | □ Verified | Installed and verified on Samsung Galaxy SM-M176B |
| **Branding & Identity** | App Icon & Launcher Label | □ Complete | Gold candlestick icon & "Trade Track" label |

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# 4. Functional Requirements Specification

## 4.1 Module 1: Splash & Initial Onboarding

- **FR-1.1 Splash Screen**:
- Displays high-resolution luxury emblem, bull & bear artwork, and brand typography.
- Inspects `hasCompletedOnboarding` state from persistent settings to route to `/dashboard` or `/onboarding`.

- **FR-1.2 Onboarding Workflow**:
- Educational feature carousels highlighting trade logging, performance analytics, and local privacy.
- Initial configuration of **Default Currency** (₹ INR, $ USD, € EUR, £ GBP).
- Initial configuration of **Default Market** (NSE India or Forex).
- Setting **Initial Starting Capital** with instantaneous validation.

## 4.2 Module 2: Executive Dashboard

- **FR-2.1 KPI Metric Cards (2×2 Grid)**:
- **Net P&L**: Color-coded (+ green / - red) with dynamic return percentage badge.
- **Win Rate**: Percentage of profitable closed trades.
- **Profit Factor**: Gross Profit divided by Gross Loss (displays `∞` when 0 losses exist).
- **Total Trades**: Total number of completed orders.
- **FR-2.2 Cumulative Equity Curve**:
- Dynamic interactive line chart plotting cumulative net P&L progression over chronologically sequenced trades.
- Visual baseline markers, gradient fills, and custom touch tooltips showing date and realized value.
- **FR-2.3 Daily P&L Histogram**:
- Bar chart summarizing individual daily net outcomes across the active filter window.
- **FR-2.4 Date Range Filter**:
- Instantaneous switching between *Today*, *This Week*, *This Month*, *Last 30 Days*, *Last 3 Months*, *This Year*, *All Time*, and *Custom Date Range Picker*.

## 4.3 Module 3: Trade Journal & Multi-Filtering

- **FR-3.1 Real-Time Search**: Sub-string filtering across symbol, setup description, strategy, and notes.
- **FR-3.2 Quick Market Filters**: Rapid toggle pills (*All*, *NSE*, *Forex*).
- **FR-3.3 Advanced Filter Bottom Sheet**:
- Filter by Strategy (multi-select).
- Filter by Outcome (*Win*, *Loss*, *Breakeven*, *Open*).
- Filter by Trade Status (*Open*, *Closed*).
- **FR-3.4 Trade Card Component**:
- Symbol & Market badge avatar.
- Date and execution timestamp.
- Net P&L with sign-prefixed currency formatting.
- Planned R:R and realized R-Multiple tags.
- Rule adherence indicator (*Rule Followed* vs *Rule Broken*).

## 4.4 Module 4: Order Entry & Real-Time Calculation Engine

- **FR-4.1 Core Parameters**:
- Symbol input with auto-capitalization.
- Market selector (*NSE*, *Forex*).
- Trade Direction toggle (*Buy / Long* vs *Sell / Short*).
- Trade Status toggle (*Closed* vs *Open*).
- Entry Price, Exit Price (optional if open), Quantity, and Contract Multiplier.
- **FR-4.2 Automated Real-Time Math**:

$$\text{Gross P\&L} = (\text{Exit Price} - \text{Entry Price}) \times \text{Quantity} \times \text{Multiplier} \quad (\text{for Long})$$ $$\text{Gross P\&L} = (\text{Entry Price} - \text{Exit Price}) \times \text{Quantity} \times \text{Multiplier} \quad (\text{for Short})$$ $$\text{Total Costs} = \text{Brokerage} + \text{Exchange Fees} + \text{Spread} + \text{Other}$$ $$\text{Net P\&L} = \text{Gross P\&L} - \text{Total Costs}$$ $$\text{Planned Risk} = |\text{Entry Price} - \text{Stop Loss}| \times \text{Quantity}

\times \text{Multiplier}$$ $$\text{Planned Reward} = |\text{Take Profit} - \text{Entry Price}| \times \text{Quantity} \times \text{Multiplier}$$ $$\text{Risk to Reward Ratio} = \frac{\text{Planned Reward}}{\text{Planned Risk}}$$ $$\text{R-Multiple} = \frac{\text{Net P\&L}}{\text{Planned Risk}}$$

- **FR-4.3 Psychological & Behavioral Logging**:
- Pre-trade and post-trade emotional state selector.
- Rule Followed compliance toggle.
- Mistake categorization and lessons learned inputs.
- **FR-4.4 Visual Attachments**:
- Gallery and camera chart screenshot picker with local image caching and preview thumbnails.

## 4.5 Module 5: Trade Details & Order Management

- **FR-5.1 Detailed Execution Breakdown**:
- Full financial recap: Entry, Exit, Net P&L, Gross P&L, Total Costs breakdown.
- Planned parameters vs realized performance.
- **FR-5.2 Psychological Review Card**:
- Emotions, discipline adherence, mistake reflections.
- **FR-5.3 Fullscreen Chart Viewer**:
- Pinch-to-zoom chart viewer for attached setup screenshots.
- **FR-5.4 Actions**:
- Modify Trade (pre-populates form) and Delete Trade with confirmation dialog.

## 4.6 Module 6: Advanced Analytics

- **FR-6.1 Timeframe Window**: 1M, 3M, 6M, 1Y, and All-Time performance aggregation.
- **FR-6.2 Win/Loss/Breakeven Distribution**:
- Donut chart with center trade counter and proportional breakdown of Wins, Losses, and Breakeven trades.
- **FR-6.3 Strategy Leaderboard**:
- Strategy table sorting strategies by Net P&L, Win Rate, and Trade Count.
- **FR-6.4 Advanced Performance Statistics**:
- Average Winner ($)
- Average Loser ($)
- Expectancy ($/trade)
- Maximum Drawdown ($ and %)
- Profit Factor

## 4.7 Module 7: Daily Review & Calendar Heatmap

- **FR-7.1 Calendar Heatmap**:
- Month calendar grid with color-coded day cells: Green for net profitable days, Red for net loss days, Gray for inactive days.
- **FR-7.2 Daily Journaling**:
- 1–5 Star Discipline Score rating.
- Text reflections: What went well, What mistakes occurred, Key takeaways.
- **FR-7.3 Execution Listing**:
- Filtered list of all individual orders executed on the selected date.

## 4.8 Module 8: Strategy Manager & Data Governance

- **FR-8.1 Strategy CRUD**: Create, edit, archive, and delete custom strategies with unique hex colors and target markets.
- **FR-8.2 CSV Export**: Export trades to RFC 4180-compliant CSV files.

- **FR-8.3 JSON Backup & Restore**: Full database backup generation and file-picker-based restore with conflict handling.
- **FR-8.4 Reseed & Reset**: Ability to wipe data or reseed high-quality demo data for demonstration purposes.

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# 5. Non-Functional Requirements (NFR)

## 5.1 Performance & Responsiveness

- **Rendering**: Consistent 60 FPS / 120 FPS frame rate utilizing Impeller Vulkan engine on Android.
- **Cold Start Time**: Launch to interactive UI under 1.2 seconds on standard mobile hardware.
- **Database Query Latency**: Less than 15ms for journal querying and aggregations up to 5,000 trades.

## 5.2 Security & Data Privacy

- **Local Isolation**: All user financial records, journals, and chart images reside strictly on the local device sandbox.
- **No Third-Party Telemetry**: Zero analytics SDKs, ad networks, or cloud trackers.

## 5.3 Reliability & Fault Tolerance

- **Database Atomicity**: ACID-compliant transactions via SQLite/Drift for multi-table operations.
- **Error Boundaries**: Try-catch encapsulated bootstrap sequence guaranteeing app display even in edge corruption cases.

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# 6. Database Entity Relationship Model

```
erDiagram
    TradesTable {
        text id PK
        datetime tradeDate
        text symbol
        text market
        text direction
        real entryPrice
        real exitPrice
        real quantity
        real contractMultiplier
        text strategy
        text setup
        real stopLoss
        real takeProfit
        real brokerage
        real exchangeFees
        real totalCosts
        real grossPnL
        real netPnL
        real riskRewardRatio
        real rMultiple
        text status
        text outcome
        bool ruleFollowed
        text screenshotPath
    }

    StrategiesTable {
        text id PK
        text name
        text description
        text colorHex
        text market
        bool isArchived
    }

    DailyReviewsTable {
        text id PK
        datetime reviewDate
        text reflection
        int disciplineScore
        text mistakes
        text lessonsLearned
    }

    AppSettingsTable {
        text key PK
        text value
    }

    StrategiesTable ||--o{ TradesTable : categorizes
    DailyReviewsTable ||--o{ TradesTable : reviews_day
```

# 7. Verification & Test Suite Summary

The application includes an automated test suite across four test tiers:

1. **Calculation Engine Tests** (`test/trading_calculation_engine_test.dart`):
- Verifies Long/Short gross & net P&L calculations.
- Verifies fee summation and breakeven threshold logic.
- Validates planned risk, reward, R:R ratio, and R-Multiple.
- Tests expectancy, win-rate, and max drawdown algorithms.
- Tests edge cases: 0 quantity, missing exit prices, zero planned risk.
2. **Database Integration Tests** (`test/database_test.dart`):
- In-memory Drift SQLite testing for trade insertion, retrieval, update, and deletion.
- Strategy CRUD and daily review date querying.

- Batch operations and database reset functionality.

3. **Formatters & Utilities Tests** (`test/utils_test.dart`):

- Currency formatting with Indian numbering format (Cr, L, K) and international currencies.
- DateTime formatting and day-boundary computations.

4. **Widget UI Tests** (`test/widget_test.dart`):

- Interactive gold button tap and rendering tests.
- Financial metric card value, sign, and badge rendering tests.
- Status badge styling across all trade outcome states.